ALLWINS KANNAN RAJAIAH

AI SOLUTIONS ARCHITECT

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PROFESSIONAL SUMMARY

- **AI Solutions Architect with 20+ years of experience** in delivering enterprise applications, cloud-native platforms, and AI-enabled solutions, translating business challenges into scalable, secure, and production-ready architectures.
- **Design end-to-end AI and cloud architectures on AWS**, integrating Generative AI, Agentic AI, Java, Spring Boot, Angular, React, microservices, REST APIs, and enterprise systems to deliver business-aligned solutions.
- **Build agentic AI solutions** using Amazon Bedrock AgentCore, Strands Agents, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), and human-in-the-loop controls, incorporating secure tool integration, observability, and evaluation
- **Lead cloud-native application architecture and AI-enabled data engineering**, combining enterprise application modernization with Databricks, Auto Loader, Lakeflow, PySpark, SQL, dbt, and Unity Catalog to deliver scalable data platforms, governed data products, automated reconciliation, and AI-powered exception investigation.
- **Drive technology strategy and engineering excellence** through governance, solution design reviews, technology roadmaps, security standards, and cross-functional leadership of development and QA teams, using AI to improve engineering productivity and operational outcomes.

CORE TECHNOLOGY

Architecture & Leadership	Solution Architecture, Cloud Architecture, Architecture Governance, Design Reviews, Enterprise Modernization, Team Leadership, Estimation & Release Readiness
AI & GenAI	Generative AI, Agentic AI, Amazon Bedrock, AgentCore, Strands Agents, MCP, RAG, Bedrock Knowledge Bases, LLM Applications, AI Governance, AI Evaluation, Langfuse
Cloud & Platform	AWS, Docker, Kubernetes, Terraform, Serverless, Cloud Security, Scalability & Cost Optimization
Data Engineering	Databricks, Lakeflow, Auto Loader, Databricks Workflows, Unity Catalog, PySpark, dbt, SQL, Data Pipelines
Application & Integration	Java, Spring Boot, Hibernate, Angular, Microservices, REST APIs, Redis, OAuth, JWT, NodeJs, HTML, CSS, JavaScript, Typescript
DevOps & Observability	CI/CD, Git, Jenkins, AWS CodePipeline, Prometheus, Splunk, Datadog, OpenTelemetry
Databases	Oracle, MySQL, DB2, Amazon RDS
AI-Assisted Development	Claude Code, GitHub Copilot, Replit, Cursor

EDUCATION

B.E., Computer Science & Engineering, Bharathiar University, India

CERTIFICATION

AWS Certified AI Practitioner **AIF-C01**
Hashicorp Certified Terraform Associate **(004)**

PROFESSIONAL EXPERIENCE

Tekizma, India. Sep 2024 – Present **Director of Technology | AI Solutions Architect**

- Lead enterprise technology direction **by defining solution architecture, target-state roadmaps, and design standards for application, data, cloud, and AI platforms, while overseeing architecture governance across development and QA teams**

Enterprise Application Modernization

- **Led the modernization of a legacy Spring/JSP enterprise application** into a cloud-native platform built on Spring Boot, Angular, REST APIs, Hibernate, MySQL, and Redis, delivered with Docker containerization, Terraform-based infrastructure provisioning, AWS CodePipeline CI/CD, and end-to-end observability using OpenTelemetry, Datadog, and Splunk.
- **Defined the target AWS architecture** across EC2, ECS/Fargate, and Lambda, with VPC networking, ALB/NLB, and CloudFront for delivery, secured through IAM, KMS, Secrets Manager, Cognito, and WAF, and monitored centrally via CloudWatch.
- **Developed AI-powered application capabilities using Agentic AI and Amazon Bedrock AgentCore**, enabling intelligent, task-oriented workflows integrated with enterprise application services and business processes.
- **Designed secure AI agent integration using the Model Context Protocol (MCP)**, enabling AI agents to interact with application tools, APIs, and enterprise services through controlled, standardized interfaces.
- **Implemented RAG-based AI workflows with an enterprise knowledge base**, leveraging enterprise knowledge retrieval and Claude Code to support intelligent information access, contextual responses, and AI-assisted application functionality.
- **Established AI-Driven Development Life Cycle (AIDLC) practices** across requirements analysis, solution design, implementation, testing, and validation, incorporating structured human review and quality gates to improve delivery consistency, maintainability, and engineering governance

Key achievement: Accelerated engineering productivity using Claude Code, Agentic AI, MCP, and RAG to streamline code development, enable intelligent workflows, and improve contextual access to enterprise knowledge.

Technologies: *Spring Boot, Angular, Hibernate, MySQL, Redis, Docker, Terraform, AWS CodePipeline, OpenTelemetry, Datadog, Splunk, Amazon Bedrock, AgentCore (Runtime, Gateway, Identity), Strands Agents, MCP, RAG, Bedrock Knowledge Bases, Claude Code*

Databricks Data Engineering & Reconciliation

- **Built a scalable Databricks data engineering platform** supporting incremental data ingestion through Auto Loader and Lakeflow and data modeling using PySpark, SQL, and dbt, and workflow orchestration through Databricks Workflows.
- **Established data governance and reusable data-product capabilities** using Unity Catalog, enabling governed data access and natural-language interaction with enterprise data, while leveraging GitHub Copilot to accelerate engineering productivity.
- **Designed the end-to-end data reconciliation architecture**, defining source-to-target mappings, data quality and validation rules, exception-handling mechanisms, and reconciliation controls across ingestion and transformation layers; implemented continuous delivery workflows using GitHub Actions.
- **Developed an AI reconciliation agent** to investigate Databricks pipeline exceptions, correlate failures with source-to-target mappings and operational runbooks and recommend corrective actions for analyst review and approval. Integrated Langfuse for execution tracing and evaluation.

Key Achievement: Improved reconciliation accuracy by 50% through automated validation, standardized exception handling, and AI-assisted exception investigation

Technologies: *Databricks, Auto Loader, Delta Live Tables, Databricks Workflows, Unity Catalog, PySpark, SQL, dbt, Langfuse, GitHub Actions, GitHub Copilot*

Advanced Software Design, Inc, USA Sep 2013 – Aug 2024 **Solutions Architect**

- **Delivered multi-tier enterprise applications** using Spring Boot, Angular, microservices, REST APIs, and AWS, with secure, high-performance APIs covering authentication, authorization, and operational readiness.
- **Designed AWS deployments** on EC2, ECS, Lambda, and Amazon RDS, using VPC, ELB, IAM, S3, and CloudFront for networking, load balancing, access control, storage, and content delivery.
- **Defined service boundaries, API contracts, and integration patterns** for complex enterprise workflows.
- **Directed application modernization**, assessing legacy components and defining incremental migration strategies.
- **Introduced reusable design patterns and technical components**, improving consistency and development efficiency.
- **Collaborated with development, QA, business analysis, and delivery teams** on estimation, release readiness, and resolution of complex production issues.

Key achievement: Delivered the official **Health Plan Marketplace for New York State of Health** as Solutions Architect, supporting **5 million enrollments in 2020** at a peak load of concurrent users

Technologies: *Spring Boot, Angular, REST APIs, Microservices, AWS (EC2, ECS, Lambda, S3, RDS, VPC, ELB, IAM, CloudFront), Hibernate, Oracle, DB2, Docker, Kubernetes, Jenkins, Git*

Marlabs Software Pvt. Ltd. Sep 2010 – Sep 2013 **Technical Manager | Mobile Solutions Architecture**

- **Defined cross-platform mobile solution architecture** for iOS, Android, and PhoneGap, covering security, authentication, REST APIs, and logging, and produced **application designs, SDLC plans, and delivery roadmaps**.
- **Delivered eFleetPro enterprise integration solutions** connecting construction-equipment manufacturers, dealers, and rental fleets with **ERP systems, punch-out integrations, and payment gateways**.
- **Established signed mobile-to-enterprise integration patterns** for secure communication with backend services and external systems.
- **Led the Mobile Centre of Excellence**, setting platform standards and managing **team formation, hiring, and delivery** across client and internal initiatives.
- **Presented cross-platform capabilities and technology solutions** to customers and leadership, supporting client engagement and revenue growth.

Technologies: *iOS, Android, PhoneGap, REST APIs, Java, Mobile Architecture, Enterprise Integration, ERP Integration, Payment Gateway Integration, Security & Authentication*

ADDITIONAL EXPERIENCE

Tech Lead	Mapview Geographical Solutions	Sep 2009 – Aug 2010
Tech Lead / Senior J2EE	Star4ce Technologies, Inc.	Nov 2007 – Jun 2009
Team Lead	Amtex Systems, Inc.	Apr 2007 – Oct 2007
Team Lead	3i-Infotech	Apr 2004 – Feb 2007
Java Developer	Asic Infotech Pvt. Ltd.	May 2003 – Feb 2004